This assignment covers our class sessions on October 4, 11, 13, and 15. Consistent with the syllabus, you'll turn in the "B" and "C" problems. The B problems will be graded out of three points each, while the C problems will be graded out of five points each.

For problems from the book, the problem numbers are from the sixth edition.

Reading

Sections 14.3–14.7. [You can't really turn this in, but I highly recommend it. I try to make sure I touch upon everything between the work we do in class and the assigned problems, but some things might only be covered obliquely, and seeing it written out can only help.]

A Practice problems

My suggestion is that you look at these problems and only attempt the ones that you're unsure how to start. If there are problems here you're unsure how to do, bring them by the MSSC or talk to me during office hours about them!

§14.3	3, 7, 18, 28, 32
§14.4	7, 11, 13, 19, 51, 55, 67, 76, 78, 83
§14.6	6, 10, 11, 26, 33, 50
§14.7	15, 19, 33

B Baseline problems

I remind you of a couple of items from the syllabus:

- My preference is that you work with other people, but please write up your solutions yourself. Don't make me wonder whether your homework was plagiarized. See the syllabus for more information. The best way to avoid getting in trouble is to write out a bit of your reasoning. To goad you into doing this, you will not receive full credit if your response to a problem is just the answer.

- A point of emphasis this semester will be on communicating your answers well. To that end, please don't just write down the answers, say something about how you decided what the answers were. Also I'd prefer if you did not turn in the paper on which you figured out the answer for the first time—organize your thoughts and then write down the organized solutions.

B1. §14.4, #54.

B2. §14.6, #18.

B3. §14.7, #44.

B4. Consider a sphere of radius 3 centered at the origin. The plane tangent to the sphere at $(1, 2, 2)$ intersects the x-axis at a point P. Find the coordinates of P.

C More involved problems

C1. In this problem we'll justify the formula we've been using that says that directional derivatives come from dot products against the gradient. The plan is to work from the limit definition of the directional derivative and try to massage it into something that looks like the dot product we think we're supposed to get. Generalizing the limit definition for the partial derivatives, the directional derivative of a function $f(x, y)$ at the point (x_0, y_0) in the direction of the unit vector $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ can be defined as

$$(D_{\mathbf{u}}f)(x_0, y_0) = \lim_{h\to 0} \frac{f(x_0 + hu_1, y_0 + hu_2) - f(x_0, y_0)}{h}.$$

Recall that the "slope" between two points on a graph is the change in z-value divided by the distance traveled in the domain; in this case, the ratio in the limit represents the slope when traveling from (x_0, y_0) a distance of h units in the direction of the vector $\mathbf{u}$. When h goes to zero, we get the derivative in that direction.

The issue here is that both x and y are changing simultaneously. The way we're going to overcome this is by replacing f with the first terms of its Taylor series! (Look more excited.) We haven't discussed Taylor series for functions of multiple variables, but we have seen the expression for the tangent plane, namely

$$f(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + \cdots.$$

Okay, to be fair, the tangent plane has no $\cdots$ at the end. But it turns out that the formula continues with terms that have $(x - x_0)$ and/or $(y - y_0)$ to higher powers than the first—we'll talk about this in next week's assignment.

(a) Explain why all the "higher order" terms have at least an h^2 in them when you plug in $x = x_0 + hu_1$ and $y = y_0 + hu_2$.

(b) Replace $f(x_0 + hu_1, y_0 + hu_2)$ with the Taylor series equation above and simplify the limit.

(c) Argue that the limit tells us that $D_{\mathbf{u}}f(x_0, y_0) = u_1 f_x(x_0, y_0) + u_2 f_y(x_0, y_0)$.

C2. §14.6, #20.

C3. In this problem we're going to look at the "standard" example of a function whose mixed partial derivatives aren't equal. Don't get your hopes too high—the function we're going to look at is piecewise-defined, meaning that when $(x, y) \neq (0, 0)$ it's a regular, run-of-the-mill rational function. Therefore the only point of contention is the origin. The mixed partials are equal everywhere else.

$$f(x,y) = \begin{cases} \dfrac{xy^3 - x^3y}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

(a) Calculate the first partial derivatives f_x and f_y at points that are different from $(0, 0)$. (For this you use the quotient rule.)

(b) Calculate $f_x(0, 0)$ and $f_y(0, 0)$. Since the function is piecewise-defined you need to do this with the limit definition of the partial derivatives. On the plus side, the limit shouldn't be too bad. (Both should come out zero.)

(c) Now I claim that the two numbers $f_{xy}(0, 0)$ and $f_{yx}(0, 0)$ don't match. Unfortunately, to check this we need more derivatives at the origin, which will require two more limits. Do those limits and check the claim.

(d) Don't write anything for this part, but if you have the opportunity you should graph this function in Maple and see if you can detect any oddity near the origin. I don't feel like it's *that* crazy. Don't expect to see wild rips or tears—the function is continuous at the origin and has continuous first partial derivatives, so it can't be that bad. Actually, if you have Maple open, plot f_x near the origin. You should see an absolute value-like thing which suggests that the second derivative could be discontinuous.

C4. One thing the gradient can do that we never discussed (until now!) is determine the vector normal to the graph of a function. This is definitely something you should know how to do in the future.

(a) Let $z = f(x, y)$ be the graph of a function. Determine a vector normal to the graph at the point $(x_0, y_0, f(x_0, y_0))$. [*Hint.* You should look back at HW#2, problem C1.]